

Q1. Write a Python program to create a DataFrame containing the attributes City, House_Type, House_Area, Number_of_Rooms, and Monthly_Rent. Apply the Normalization technique to the numerical attributes and display the normalized data.

[10 Marks]

Q2. Generate a DataFrame containing customer information for Netflix and Amazon Prime users. Create a Venn diagram and a Pie chart to show the customers using only Netflix, only Amazon Prime, and both services. Add appropriate labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Apply the K-Means Clustering algorithm to a student dataset containing Attendance and Marks. Perform the necessary data preprocessing, use $K = 3$ to group the students into three clusters, and visualize the resulting clusters along with their cluster centroids using a scatter plot.

[20 Marks]

Q3. Viva

[5 Marks]